

Linear Motion · Practice Problem Set 1 (Solution)

Problem 1.

Given: initial velocity $v_0 = 0 \text{ m/s}$, acceleration $a = 2.0 \text{ m/s}^2$, time $t = 5.0 \text{ s}$.

(a) $v = v_0 + a \cdot t$

Substituting the given values, v equals 10.0 m/s .

(b) $d = v_0 \cdot t + \frac{1}{2} \cdot a \cdot t^2$

Substituting the given values, d equals 25.0 m .

The cyclist reaches 10.0 m/s and travels 25.0 m .